MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

DEPARTMENT OF COMPUTER ENGINEERING

SEMINAR INTRODUCTION REPORT (K-SCHEME — SEMESTER V)

Seminar Topic: Agentic AI: The Evolution from Chatbots to Autonomous AI Systems	**Group No:** 01
Major Technologies: Generative AI, AI Agents, LLMs	**Academic Year:** 2026–2027
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1. Introduction

Artificial Intelligence has evolved from basic rule-based systems to conversational chatbots, and now to Agentic AI. Unlike traditional chatbots that only answer questions when prompted, Agentic AI refers to autonomous systems that can perceive their environment, think, plan sequential steps, make decisions, and take actions using external digital tools to accomplish complex goals with minimal human supervision.

2. Background & Need for the Technology

- **Evolution:** First-generation chatbots provided static, scripted responses. Generative AI models (e.g., ChatGPT) introduced human-like text synthesis. Agentic AI represents the next logical leap by bridging language understanding with autonomous action and tool execution.
- **The Problem:** Traditional chatbots are passive—they cannot independently complete multi-step tasks like browsing the web, updating databases, writing and executing code, or scheduling workflows without continuous human prompts.
- **The Need:** Modern industries require proactive systems capable of executing end-to-end tasks autonomously, reducing human workload, eliminating manual bottlenecks, and operating 24/7.

3. Seminar Objectives

- **Objective 1:** To understand the fundamental architectural shift from passive conversational chatbots to goal-driven autonomous agents.
- **Objective 2:** To study the core working cycle of Agentic AI: Perception, Planning, Memory Management, and Tool Integration.
- **Objective 3:** To evaluate real-world industrial use cases, current challenges, technical limitations, and career scope for Computer Engineering diploma students.

4. Basic Concepts & Fundamentals

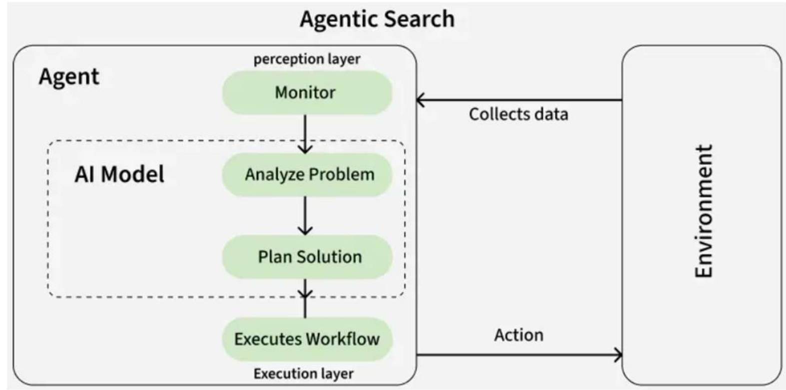
- **AI Agent:** An autonomous software entity that senses its environment and takes actions to achieve a predefined objective.
- **Planning & Reasoning:** The ability of an LLM brain to decompose high-level instructions into smaller logical sub-tasks.
- **Memory Systems:** Short-term memory retains context within the current session, while long-term memory retrieves knowledge across past interactions.
- **Tool / Action Interface:** Mechanisms enabling the AI to interact with external systems such as web search, APIs, file systems, and Python interpreters.

5. Working Architecture & Mechanism

Agentic AI operates in a continuous, iterative feedback loop (Perceive -> Plan -> Act -> Observe):

- **1. User Goal:** The user provides an objective (e.g., 'Analyze sales data from the server and generate a PDF summary').
- **2. LLM Core (Brain):** The model analyzes the prompt, queries its memory, and establishes a sequential plan of execution.
- **3. Tool Invocation:** The agent triggers external APIs, runs scripts, or queries databases to perform necessary actions.
- **4. Observation & Self-Correction:** The agent monitors tool outputs. If errors or unexpected outputs occur, it refines its plan and re-executes until the final goal is met.

6. Key Industry Applications



- **Customer Support & Service:** Autonomous agents handle complex tickets end-to-end, issuing refunds, updating CRM records, and modifying accounts.
- **Automated Software Engineering:** AI agents inspect codebases, write unit tests, detect bugs, and push pull requests autonomously.
- **Enterprise Workflow Automation:** Parsing documents, cross-referencing spreadsheets, drafting financial memos, and scheduling meetings without human intervention.

7. Advantages vs. Technical Limitations

- **Advantages:** High efficiency, continuous 24/7 availability, handles complex multi-step workflows, and significantly reduces manual overhead.
- **Limitations:** Higher computational cost, risk of infinite loops or hallucinated tool parameters, and the need for strict security guardrails.

8. Skills Required & Career Opportunities

- **Industry Skills:** Python programming, understanding of REST APIs, foundational LLM concepts, prompt engineering, and basic vector database usage.
- **Career Opportunities:** AI Application Developer, Automation Engineer, Generative AI Specialist, and Python Systems Developer.

9. References

- **[1]** MSBTE K-Scheme Curriculum Guidelines: Emerging Technologies & Seminar (CO Branch).
- **[2]** Wang, L. et al., 'A Survey on Large Language Model based Autonomous Agents', Frontiers of Computer Science, 2023.
- **[3]** OpenAI & LangChain Technical Documentation on Autonomous Function Calling and Agent Architectures.